# Rubric 6 Home Lab Guide

## Configure Cryptographic Protocols on hvmc-dc01 (TLS · SSH keys · LDAPS · Wireshark)

*A beginner's step-by-step guide for the home MacBook Air. About 2 hours. Every script explains what it does, saves its output for the report, and tells you when to take a screenshot.*

| | |
|---|---|
| **Project** | Healthcare IAM Operations Simulation — Horizon Valley Medical Center (HVMC) |
| **Rubric** | Cryptography 1 — Configure Cryptographic Protocols (rubric 6) |
| **Prepared By** | Sabitha S Karanavar |
| **Version** | 1.1 — November 3, 2026 (first version: home kit guide, October 26, 2026) |
| **Where** | Home MacBook Air (VirtualBox) and the domain controller VM `hvmc-dc01` (192.168.56.10) |
| **Cost** | $0 — everything runs on your own Mac and VM; no Azure |

| What you will build | Rubric requirement |
|---|---|
| HTTPS on the domain controller's web server (nginx) with a certificate from **your own HVMC PKI**, TLS 1.2/1.3 only | Configuration of TLS for a web server |
| SSH login with an Ed25519 key; password and root logins switched off | Setup of SSH with key-based authentication |
| LDAPS: the directory (Samba AD) talks over TLS with the HVMC certificate; plain-LDAP password logins refused | Implementation of a secure communication channel |
| Captures of HTTP vs HTTPS, LDAP vs LDAPS and SSH, analysed with tshark/Wireshark | Analysis of protocol security using Wireshark |

## Part 0 — Before you leave the office

This guide is on GitHub, so you can open it at home. The **kit itself is not on GitHub**, because it holds the server's private key. Bring the kit home yourself:

| # | Do this at the office | Why |
|---|---|---|
| 1 | Email `HVMC_Rubric6_Home_Kit.zip` (in **Downloads** on the office Mac) to yourself as an attachment | The kit holds the scripts and the server certificate and key |

| # | Do this at the office | Why |
|---|---|---|
| 2 | Make sure you know the **kit password** (given to you on 26 Oct; it is not written down anywhere) | `unzip` asks for it in Step 3. If you do not have it, ask Claude to rebuild the kit with a new password *before* you go home |
| 3 | At home, open this guide: GitHub → `cryptography_1/crypto_protocols/HVMC_Rubric6_Home_Lab_Guide.pdf` | Keep it open next to Terminal |

**What you need at home:** the home MacBook Air with VirtualBox, the **HVMC-DC01** VM, your VM login name and password (Week 1 document), internet for the VM (to install packages), and about **2 hours**. The client VM **HVMC-CLIENT01** is not needed — leave it switched off.

## What is in the kit

| Folder | Contents |
|---|---|
| `certs/` | HVMC Root CA R1 and Issuing CA 1 certificates, the server certificate for `hvmc-dc01.hvmc.lan` (also `portal.hvmc.lan` and 192.168.56.10), and its private key. **Lab-only key** made for this exercise. |
| `mac/` | Scripts you run in the Mac's Terminal: `m1` check · `m2` copy kit to VM · `m3` SSH keys · `m4` test traffic · `m5` fetch evidence |
| `vm/` | Scripts you run on the VM: `00` precheck · `01` install · `02` HTTPS · `03` LDAPS · `04` SSH hardening · `05` capture · `06` analyse · `07` collect |

> **Safety rules for this lab.** (1) Take the VirtualBox snapshot in Step 2 first — if anything goes wrong, you restore it and nothing is lost. (2) In Step 10 keep your VM window open until Step 11 says OK. (3) The only "password" that appears in the captures is the made-up `NotARealPassword-Demo` / `Fake-Demo-Password-1`. Never type your real passwords into the test commands. (4) Do not put the kit's `.key.pem` file on GitHub.

## Part A — Get ready (15 minutes)

### Mac Step 1 — Check whether your Mac account is an administrator

Two ways: **(a)** Apple menu → System Preferences → Users & Groups: under your name it says **Admin** or **Standard**. **(b)** The script in Step 4 also tells you. Either answer is fine — it only changes how you install Wireshark (Step 18).

### VirtualBox Step 2 — Take a safety snapshot

In VirtualBox: if HVMC-DC01 is running, shut it down cleanly first (in the VM: `sudo poweroff`). Select HVMC-DC01 → **Snapshots** → **Take** → name it **before rubric 6** (keep your earlier Week 1–5 snapshots). Then start the VM again (**Start** → **Headless Start**) and wait about 1 minute.

📸 **Screenshot:** the snapshot list with the new snapshot

### MAC Terminal Step 3 — Unzip the kit

Open the email you sent yourself (Part 0) and save `HVMC_Rubric6_Home_Kit.zip` into **Downloads**. Open **Terminal** (Applications → Utilities) and type the commands below. `unzip` asks for the kit password.

```
cd ~/Downloads
unzip HVMC_Rubric6_Home_Kit.zip
cd HVMC_Rubric6_Home_Kit/mac
```

✔ **You should see:** a list of extracted files, no errors

### MAC Terminal Step 4 — Check the Mac

```
./m1_check_mac.sh
```

✔ **You should see:** your admin answer, the tools list (ssh, curl, openssl, ldapsearch found) and ✅ **SSH port 22 on 192.168.56.10 is reachable**

📸 **Screenshot:** this window

## Part B — Copy the kit and check the server (15 minutes)

MAC Terminal **Step 5 — Copy the kit to the VM**

Replace `YOURNAME` with the username you use to log in to the VM (the one in your Week 1 document, Figure 11). Type your VM password when asked.

```
./m2_copy_kit_to_vm.sh YOURNAME
```

✔ **You should see:** ✅ Kit copied

MAC Terminal **Step 6 — Open a second Terminal window for the VM**

Press ⌘**N** for a new Terminal window and log in to the VM. Keep **both** windows open: one says *MAC*, one says *VM* in this guide.

```
ssh YOURNAME@192.168.56.10
cd ~/hvmc_kit/vm
```

VM (ssh window) **Step 7 — Precheck (read-only)**

```
./00_precheck.sh
```

✔ **You should see:** ✅ Precheck passed: hvmc-dc01.hvmc.lan at 192.168.56.10, Samba running

📸 **Screenshot:** this window

VM (ssh window) **Step 8 — Install the tools**

Installs nginx (web server), ldap-utils and tshark (command-line Wireshark). Needs your VM password.

```
./01_install_packages.sh
```

✔ **You should see:** ✅ Packages installed

## Part C — SSH with key-based authentication (20 minutes)

MAC Terminal **Step 9 — Create your SSH key and install it**

Creates `~/.ssh/hvmc_lab_ed25519` on the Mac. It asks you to choose a **passphrase** for the key (remember it), then your VM password one last time. From then on you log in with `ssh hvmc-dc01`.

```
./m3_ssh_key_setup.sh
```

✔ **You should see:** ✅ Key login works

📸 **Screenshot:** the key fingerprint and the 'key login OK' line

### VM (ssh window) Step 10 — Switch off password logins

Log out of the VM window (`exit`) and log in again with the key: `ssh hvmc-dc01`, then `cd ~/hvmc_kit/vm`. Run the hardening script. **Keep this window open** until Step 11 is green.

```
./04_ssh_hardening.sh
```

✔ **You should see:** BEFORE (passwordauthentication yes) and AFTER (passwordauthentication no, permitrootlogin no) and ✅ SSH now accepts keys only

📸 **Screenshot:** the BEFORE and AFTER settings

### MAC Terminal Step 11 — Prove it

Checks that your key still works and that password, root and old-cipher logins are refused.

```
./m3_ssh_key_setup.sh verify
```

✔ **You should see:** three 'Permission denied' / 'no matching cipher' results and ✅ Verified

📸 **Screenshot:** all five results

## Part D — TLS for the web server (15 minutes)

### VM (ssh window) Step 12 — HTTPS with the HVMC certificate

Installs the certificate and key (key readable by root only), a small fictional patient-portal page, and an nginx configuration that allows only TLS 1.2 and 1.3. For the Wireshark comparison, plain HTTP still works until Step 17.

```
./02_tls_nginx.sh setup
```

✔ **You should see:** Verify return code: 0 (ok), Protocol TLSv1.3, and TLS 1.1 refused with an alert

📸 **Screenshot:** the 'certificate chain and handshake' and 'TLS 1.1 must be REFUSED' output

## Part E — Secure channel: LDAPS on the domain controller (15 minutes)

### VM (ssh window) Step 13 — LDAPS with the HVMC certificate

Shows the certificate Samba uses today (self-signed, not trusted), installs the HVMC one, restarts Samba, and tests. smb.conf is backed up first and restored automatically if Samba does not start. At the end it offers an optional count of your 30 staff accounts over LDAPS (asks for the lab administrator password, which travels encrypted).

```
./03_ldaps_samba.sh
```

✔ **You should see:** BEFORE: verify error; AFTER: issuer HVMC Issuing CA 1, Verify return code: 0 (ok); plain-LDAP bind refused (*Strong(er) authentication required*); LDAPS bind reaches the server (*Invalid credentials* for the fake password)

📸 **Screenshot:** the BEFORE and AFTER certificate output and the two bind results

# Part F — Wireshark analysis (30 minutes)

For the first three captures you use **both** windows: start the capture in the VM window, run the traffic in the MAC window, then press Enter in the VM window.

### VM (ssh window) Step 14 — Capture 1: plain HTTP

VM window: `./05_capture.sh http` — it waits. MAC window: `./m4_traffic.sh http`. Back in the VM window press Enter.

```
./05_capture.sh http          # VM
./m4_traffic.sh http          # MAC, while the VM waits
```

✔ **You should see:** ✅ Captured N packets into pcaps/http.pcap

### VM (ssh window) Step 15 — Captures 2 and 3: HTTPS and SSH

Same pattern.

```
./05_capture.sh https   +   ./m4_traffic.sh https
./05_capture.sh ssh     +   ./m4_traffic.sh ssh
```

✔ **You should see:** two more ✅ Captured lines. In the HTTPS traffic, the third test fails on purpose: without the HVMC root the Mac does not trust the server

### VM (ssh window) Step 16 — Captures 4 and 5 (automatic) and the analysis

The LDAP captures run by themselves on the VM. Then tshark analyses all five.

```
./05_capture.sh ldap
./05_capture.sh ldaps
./06_analyze.sh
```

✔ **You should see:** a summary like: NotARealPassword-Demo found in http.pcap: 1 time(s) — in https.pcap: 0 time(s); Fake-Demo-Password-1 in ldap.pcap: 1 — in ldaps.pcap: 0

📸 **Screenshot:** the summary lines

### VM (ssh window) Step 17 — Harden the web server

Now that the comparison is captured, plain HTTP only redirects to HTTPS and HSTS is switched on.

```
./02_tls_nginx.sh harden
```

✔ **You should see:** HTTP/1.1 301 Moved Permanently and a Strict-Transport-Security header

📸 **Screenshot:** the redirect and HSTS output

# Part G — Collect, Wireshark screenshots, finish (20 minutes)

### Mac Step 18 — Install Wireshark (optional but recommended)

Download the macOS installer from **wireshark.org/download.html**. Open the .dmg and drag Wireshark to **Applications** (admin account) or to an **Applications** folder inside your home folder (standard account — create it in Finder if needed). If the newest version will not open on Big Sur, choose a 4.2.x version from "Older Releases". You only *open* capture files, so no capture permissions are needed.

### VM (ssh window) Step 19 — Collect the evidence

```
./07_collect_evidence.sh
```

✔ **You should see:** ✅ Packed ~/rubric6_evidence.tar.gz

### MAC Terminal Step 20 — Bring the evidence to the Mac

The script pauses so you can take the Wireshark screenshots below and put all screenshots into the `screenshots` folder it creates on your Desktop. Then press Enter.

```
./m5_fetch_evidence.sh
```

✔ **You should see:** ✅ Done — HVMC_Rubric6_Evidence.zip on your Desktop

| Open in Wireshark (Desktop → HVMC_Rubric6_Evidence → rubric6_evidence → pcaps) | What to do | What it shows |
|---|---|---|
| `http.pcap` | Filter `http.request.method == "POST"` → right-click the packet → Follow → HTTP Stream | The fake username, password and MRN readable by anyone on the network |
| `https.pcap` | Filter `tls` ; click the *Client Hello*, expand TLS → Handshake → Extension: server_name | Only the server name and handshake are visible; data is *Application Data* |
| `https.pcap` | Statistics → Protocol Hierarchy | Everything after the handshake is encrypted TLS |
| `ldap.pcap` | Filter `ldap` ; click the *bindRequest*, expand to *simple* | The (fake) password in clear text, then Samba's refusal |
| `ldaps.pcap` | Filter `tls` | The same conversation, encrypted |
| `ssh.pcap` | Filter `ssh` ; click the *Key Exchange Init* packets | Plain-text version banners and algorithm lists, then only encrypted packets |

### VirtualBox Step 21 — Final snapshot

Shut the VM down (`sudo poweroff` in the VM window), take snapshot **rubric 6 protocols configured**, then email `HVMC_Rubric6_Evidence.zip` from your Desktop to yourself. At the office, tell Claude: "*the rubric 6 evidence is in Downloads*".

## Screenshot checklist

Tick these off as you go. They all go into the `screenshots` folder on your Desktop in Step 20.

| ✓ | Step | Screenshot | Shows (rubric) |
|---|---|---|---|
| □ | 2 | VirtualBox snapshot list | Safe change practice |
| □ | 4 | Mac check: tools found, port 22 reachable | Lab ready |
| □ | 7 | VM precheck passed | Lab ready |
| □ | 9 | SSH key fingerprint and key login OK | SSH key-based authentication |
| □ | 10 | SSH settings BEFORE and AFTER | SSH key-based authentication |
| □ | 11 | Password, root and old-cipher logins refused | SSH key-based authentication |
| □ | 12 | Certificate chain OK, TLS 1.3, TLS 1.1 refused | TLS for a web server |
| □ | 13 | LDAPS certificate BEFORE/AFTER and the two bind results | Secure communication channel |
| □ | 16 | tshark summary: fake password seen in HTTP/LDAP, not in HTTPS/LDAPS | Wireshark analysis |
| □ | 17 | HTTP redirect and HSTS header | TLS for a web server |
| □ | 20 | Six Wireshark views from the table above | Wireshark analysis |

**When you are back at the office:** put `HVMC_Rubric6_Evidence.zip` in Downloads and tell Claude. Claude then writes the rubric report `HVMC_Cryptographic_Protocols.pdf` from your evidence and pushes it with this folder.

## If something goes wrong

| Problem | What to do |
|---|---|
| `Permission denied (publickey)` after Step 10 and the key does not work | In the VM window you kept open: `sudo rm /etc/ssh/sshd_config.d/00-hvmc-hardening.conf && sudo systemctl reload ssh`. Or use the VirtualBox window to log in. Or restore the *before rubric 6* snapshot. |
| Mac cannot reach 192.168.56.10 | Start the VM in VirtualBox and wait 1 minute; check the Week 1 host-only network (Adapter 2) is still attached. |
| Samba does not start after Step 13 | The script restores the old smb.conf automatically. Send Claude `~/rubric6_evidence/logs/03_ldaps_samba.txt`. |
| `apt-get` errors in Step 8 | The VM needs internet through Adapter 1 (NAT). Try again in a minute: `./01_install_packages.sh`. |
| A capture says it is empty | Run it again and make sure you run the Mac command *while* the VM window says it is waiting. |
| Anything else, or you are unsure | Stop, restore the *before rubric 6* snapshot if needed, and bring the log files — every script saves its output in `~/rubric6_evidence/logs`. |

## Words you will see

| Word | Meaning |
|---|---|
| TLS | The protocol behind https://. It proves the server's identity with a certificate and encrypts everything. |
| Certificate / CA | A signed ID card for a server. Your HVMC Issuing CA signed it; the Mac trusts it only when given the HVMC root. |
| SSH key | A key pair: the private key stays on your Mac, the public key goes to the server. Nothing secret is sent when you log in. |
| LDAP / LDAPS | How programs read the directory (users, groups) from the domain controller. LDAPS = LDAP inside TLS. |
| Packet capture (pcap) | A recording of network traffic. Wireshark/tshark show what an eavesdropper would see. |
| Snapshot | A saved copy of the whole VM you can go back to. |